CONTENTS

EduOS Cameroon — Master Implementation Plan

1. What we are building (the one-page system picture)

2. The two clocks: T0 and the school year (normative scheduling rule)

3. Timeline overview

4. Phase 0 — Prepare (now → contract signature)

5. Phase I — Build & pilot (T0 → T0+18m, pilot pinned per MIP-R1)

Track 1 — Software build (Opesware)

Track 2 — Data & devices (PMU + Opesware) (new; closes gaps #2, #5-part)

Track 3 — People readiness (PMU) (new; closes gaps #5, #6, #10)

Track 4 — Supply-chain season alignment (PMU + ministries) (new; closes gap #3)

6. Phases II–IV — Scale (rollout engineering)

7. The dependency spine (what blocks what) — now including the physical world

8. Gate governance and the failure protocol (closes gaps #11, #12)

9. What “done” means (end state, T0+60m)

10. v1.0 → v1.1 gap register (traceability)

EduOS Cameroon — Master Implementation Plan

Document ID	EDUOS-MIP-001
Version	1.1 — closes the 13 gaps identified in the v1.0 adversarial review (gap register in §10)
Purpose	The single picture of **what is being built, in what order, by whom, over what timeline** — from governance decree to national operation
Developer	Opesware Technologies, Douala — www.opesware.com · eudos@opesware.com
Anchors	Budget phases (BUD §1) · M&E gates (MEF §3.3) · Readiness checklist (PRC) · FRS trio (04/07/08) · ADRs (doc 09)

---

1. What we are building (the one-page system picture)

EduOS Cameroon is a national education resource operating system. Its first product is complete digital lifecycle management of textbooks — every approved title, every printed copy, every movement from press to learner's hands and back — for both ministries, across 18,500+ public schools, working offline-first because that is Cameroon's reality.

```
                          ┌ EduOS Cameroon (Phase I product) ─────────────┐
 Publishers/Printers ─▶ │ CURRICULUM &  ─▶  CUSTODY &        ─▶ SCHOOL     │ ─▶
Learners
 (batch registration)   │ CATALOGUE         LOGISTICS (NWIDMS)   OPERATIONS │    (as-
signment,
                        │ (NTR titles)      warehouses,          receipts,  │     re-
turn,
 Ministry planners    ─▶ │                   shipments,           assignment,│     con-
dition)
 (allocation plans)     │ SCHOOL REGISTRY   custody chain        campaigns  │
                        │ (NSR - foundation) ASSET PASSPORT (per-copy chain) │
                        ├─────────────────────────────────────────────────┤
                        │ Platform: IAM (Keycloak) · Sync Engine · API GW    │
                        │ Analytics & Dashboards (read-only) · Public APIs   │
                        └─────────────────────────────────────────────────┘
      One Laravel modular monolith + Flutter offline app + PostgreSQL  (ADR-01..05)
```

The five things the system guarantees (each traceable to FRS requirements):

1. One authoritative register of schools (NSID) and textbooks (NTID) — no duplicates, no ghosts.
2. Every copy/batch has a tamper-evident passport — custody is always attributable.
3. No shipment closes with unexplained variance — leakage becomes visible (attacks the measured 30% PETS-III loss).
4. Schools work fully offline for up to 90 days — remote schools sync by travelling (ADR-11 rugged tier).
5. Ministries see real coverage, real stock, real losses — and the public gets Cameroon's first open school & textbook APIs.

What is deliberately NOT in Phase I (scoped to later phases so Phase I can succeed): teacher management, inspection module, e-content/NEDIH, equipment/asset passports beyond textbooks, AI/analytics beyond standard dashboards, student registry full build (Phase I uses class-level assignment fallback, FRS FR-NTR-SM-02).

2. The two clocks: T0 and the school year (normative scheduling rule)

The plan runs on two clocks that must be reconciled explicitly:

- **T0** = financing effectiveness (sets when money and contracts exist).
- **SY** = the school year: procurement/printing ~Feb–Jun, warehouse receipt ~Jun–Aug, **distribution peak Aug–Oct**, verification campaigns ~Mar–May.

Rule MIP-R1 (school-year pinning): the pilot is only valid if it spans one full **distribution season (Aug–Oct)** plus the following **verification season (Mar–May)**. Therefore pilot go-live is pinned to **the 1st of July following readiness**, not to a T0 offset. If build readiness (increment 5, §5) lands after 1 May, pilot go-live waits for the next school year rather than entering the season half-ready — and the intervening months are used for soak testing, training completion, and brownfield stock registration. Every subsequent rollout wave (Phases II–III) is likewise pinned: **school onboarding waves complete training and devices by 30 June to be live for the Aug–Oct season.**

Rule MIP-R2 (financier-clock decoupling): Phase 0's duration is set by the appraisal process, which historically runs **12–18 months**, not 6. All Opesware-controlled Phase-0 work (§4 workstreams A–C, G) is scheduled to complete within 6 months of starting, i.e. **before** the earliest plausible T0 — so the institutional clock can never be blamed on, or hidden behind, the technical one. The plan's phase math therefore shows both: duration from T0, and the SY pin.

3. Timeline overview

```
Pre-T0 (now, 12–18m)   │ Phase I (T0 → T0+18m,       │ Phase II         │ Phase III        │
Phase IV
incl. Phase 0 work     │ pilot pinned per MIP-R1)    │ (→ T0+30m)       │ (→ T0+42m)       │
(→ T0+60m)
───────────────────────┼─────────────────────────────┼──────────────────┼──────────────────┼───
Decree · French pack   │ Build NSR→NTR→NWIDMS→       │ 5 regions        │ All 10 regions   │
Full operation
Appraisal · ADR-09/10  │ School Ops · device tender │ ~8,000 schools   │ ~18,500 schools │
2 full school
OpenAPI · DPIA · data  │ & delivery · training ToT   │ (by 30 Jun of    │ + private        │
years of stable
Procurement route      │ 500-school pilot over one   │ its SY, MIP-R1) │   onboarding     │
operation, then
Baseline studies       │ full Aug–Oct season         │                  │                  │
handover
                       │ GATE 1                      │ GATE 2           │ GATE 3           │
final evaluation
```

Phase IV ends at **T0+60m**, matching the Budget’s 5 programme years exactly (closes v1.0 gap #13). Gates are the M&E disbursement gates (MEF §3.3) with the failure protocol of §8.

4. Phase 0 — Prepare (now → contract signature)

Workstream	Deliverables	Owner	Done by
A. Contracts-as-specs	OpenAPI 3.1 files (D-NSR/NTR/NWD-API) from FRS §7; ambiguities feed FRS v1.1	Opesware	Start+2m
B. De-risk the hard parts	ADR-09 sync evaluation + prototype passing 30-day soak; ADR-10 offline-auth spec + ANTIC submission	Opesware	Start+5m
C. French pack + PDFs	fr/ versions of docs 00–11; branded PDFs for appraisal	Opesware	Start+3m (before appraisal missions)
D. Institutional	Decree (R1); financing appraisal; procurement route (PRC 1.1–1.3)	Ministries/PMU	sets T0
E. Data & legal	Carte scolaire extracts; subject codesets & curriculum versions (needed by build M4); DPIA; hosting + escrow agreements	PMU	before T0
F. Pilot prep	2 pilot regions + 500 schools selected; time-and-motion & TBD-P baseline studies **before any system touches a school**	PMU/M&E	before pilot SY
G. Device validation	Rugged reference device (Blackview Active 8 Pro class) field-tested; tender documents drafted and pre-cleared with financier so the tender can launch the day its data dependency clears (§5 track 2)	Opesware/PMU	Start+5m
H. Publisher/printer engagement *(new, closes gap #7)*	QR-at-press technical spec issued to publishers/printers; 2+ printers complete a label test run; label clauses drafted into the next print-contract template (Risk R12)	PMU + Opesware	before the pilot-year print season

Exit gate (= PRC §5): decree signed · financing effective · procurement route resolved · FRS signed off · ADR-09/10 closed · carte scolaire data in hand · DPIA done · baselines scheduled · printer label test passed.

5. Phase I — Build & pilot (T0 → T0+18m, pilot pinned per MIP-R1)

Phase I is **four parallel tracks**, not one. v1.0 showed only track 1; the other three are where national programmes actually slip.

TRACK 1 — SOFTWARE BUILD (OPESWARE)

#	Increment (months)	What ships	Why this order
1	M1–M3	**Platform skeleton**: monolith scaffold with module boundaries + CI boundary checks (Deptrac, merge-blocking), Keycloak, API gateway, sync gateway (per ADR-09 outcome), observability, dev/staging/training environments with seeded data	Everything stands on it; boundary discipline must exist before the first feature
2	M2–M5	**NSR — School Registry**: import pipeline, dedup, hierarchy, GPS verification app, enrolment returns	Foundational registry; unblocks the data campaign (track 2) which unblocks devices
3	M4–M7	**NTR — Catalogue & Passport**: titles, editions, NTID/NCID, batches, label export, public catalogue API (consumes subject codesets from Phase 0-E)	Catalogue must exist before the print season registers batches
4	M6–M10	**NWIDMS — Custody & Logistics**: stock ledger, shipments, custody chain, discrepancy cases, receipt flows	Consumes NSR (destinations) + NTR (what moves)
5	M8–M12	**School Operations (Flutter app complete)**: receive, assign, return, condition, verification campaigns — full offline cycle on both device tiers	Consumes all three registries; readiness for MIP-R1 pinning is measured here
6	M10–M13	**Dashboards & reports**: national/region/division/school views, M&E indicator feeds (OUT-1/2/5, SYS-1..4), season-readiness view	Read-only projections over now-real event streams
—	M11	**Penetration test 1 (external) — MUST pass before any learner data enters the pilot** (closes gap #9; pentest 2 pre-Phase III at national scale)	Security gate, budgeted BUD §3.1

TRACK 2 — DATA & DEVICES (PMU + OPESWARE) *(NEW; CLOSES GAPS #2, #5-PART)*

When	Activity
M3–M8	**NSR data-cleaning campaign** in pilot regions (then nationally): division validation workshops, GPS verification, tier classification. Output at M8: authoritative pilot-school list **with device-tier counts**
M8	**Device tender launches** (documents pre-cleared in Phase 0-G, so zero drafting delay)
M8–M14	Award (M10) → manufacture & shipping (M10–M13) → **customs clearance at Douala (plan 6 weeks; duty/exemption letter obtained in Phase 0-D)** → MDM enrolment, app pre-load, staging (M13–M14). **Total lead time budgeted: 6 months tender-to-school.** This line sits ON the dependency spine (§7)
M14–pilot go-live	Device distribution to pilot schools alongside training (track 3)

TRACK 3 — PEOPLE READINESS (PMU) *(NEW; CLOSES GAPS #5, #6, #10)*

When	Activity
M8	**Ministry UAT team stood up** (6–8 officers, both ministries) and trained on the FRS acceptance criteria they will witness; they co-execute increment UATs from M9 onward — not a rubber stamp at the end
M8–M9	Training materials + training environment (seeded realistic data) ready; **training-of-trainers**: 60 national trainers certified
M10–M14	Regional trainer certification (pilot regions), then **school-level training waves** synced to device distribution — 1,000 staff certified before go-live (OUT-P5 Y2 target)
M9–M13	**User provisioning campaign**: head-teacher identity proofing rides the data-campaign workshops (track 2 — same room, same trip); accounts + device-bound offline credentials (ADR-10) issued at training; target 100% of pilot schools credentialed before go-live
M12	**Support model live before the pilot, not during** (closes gap #8): L1 = division focal points (trained officers, part of BUD §3.4 scope); L2 = PMU helpdesk (2 posts within the PMU's 8, BUD §3.7, with toll-free line + WhatsApp channel); L3 = Opesware/national engineering. SLAs: L1 response same-day, L2 48h, L3 per severity. Call volumes become an M&E operational metric

TRACK 4 — SUPPLY-CHAIN SEASON ALIGNMENT (PMU + MINISTRIES) *(NEW; CLOSES GAP #3)*

When	Activity
Phase 0-H → pilot print season (Feb–Jun of pilot SY)	The **pilot-year print contract** includes QR-at-press clauses; PMU confirms with MINEDUB/PAREC that at least one national print order lands inside the pilot window. **Fallback (so the pilot never dies waiting on procurement):** if no national print run occurs in the pilot SY, the pilot substitutes (a) label-retrofit of one warehouse's existing stock plus (b) full brownfield school-stock registration (FR-NTR-MIG-02) — exercising every custody flow except at-press labelling, which is then verified in Phase II's print season. Gate 1 criteria apply to whichever path ran
Jun–Aug of pilot SY	Real (or fallback) batch: warehouse receipt → allocation → dispatch
Aug–Oct (pinned)	**PILOT DISTRIBUTION SEASON** — 500 schools receive, assign; sync at scale including travel-to-sync rugged tier
Nov–Feb	Steady-state operation, monthly reconciliation drills, support-model shakedown
Mar–May	**Verification campaign season** (FR-NTR-12) completes the full annual cycle

GATE 1 (end of pilot verification season): OUT-P3 ≥95% · OUT-1 ≥90% · one full 30-day offline cycle with zero data loss · baselines frozen in the Baseline Report · economic model re-run with measured values. Governed by §8.

Team (Opesware + national counterparts embedded from M1, Risk R4): 1 architect · 6–8 backend · 3–4 Flutter · 1–2 web · 2 QA/test-automation · 1 DevOps · 1 UX · 1 delivery lead — plus 4 national engineers as counterparts and the PMU. Definition of done everywhere = the FRS acceptance criterion passes in CI/UAT, not "demo works."

6. Phases II–IV — Scale (rollout engineering)

Every wave repeats the Phase I tracks 2–3 pattern (data-validate → devices → train+credential → go live **by 30 June**, per MIP-R1):

Phase	Window	Content	Gate
II	T0+18–30m	5 regions / ~8,000 schools; regional warehouses live; brownfield stock registration at scale (FR-NTR-MIG-02); **at-press labelling verified in this print season if the pilot used the fallback path**; device wave 2 tendered at Phase-I award prices (option clause in the wave-1 tender)	GATE 2: OUT-P2 ≥8,000 · SYS-2 ≥90%
III	T0+30–42m	All 10 regions / ~18,500 public schools (NW/SW/Far North last, emergency mode where needed, R6); private-school BYOD onboarding; student-level assignment switches on when the National Student Registry lands; **penetration test 2 at national scale**	GATE 3: national coverage · OUT-1 ≥97%
IV	T0+42–60m	Consolidation across **two full school years of stable national operation**: performance optimization, module extensions (inspection hooks, asset passports), K3s migration if ops-ready (G10), columnar analytics evaluation (G11); **operational handover to the 14-person national team** with Opesware moving to a support-contract role	Final evaluation (MEF §5); recurrent line active in national budget

Throughout II–IV the platform team shrinks as the national team grows — the crossing point is a contractual milestone, not a hope.

7. The dependency spine (what blocks what) — now including the physical world

```
Decree ─▶ Financing ─▶ Contract ─▶ Skeleton ─▶ NSR ─▶ NTR ─▶ NWIDMS ─▶ School Ops ─┐
             ▲              ▲                    │                               ├─▶
PILOT ─▶ GATE 1 ─▶ Scale
       French pack    ADR-09/10 closed         ▼                               │
(pinned to
       (appraisal)    OpenAPI, DPIA       Data campaign ─▶ Device tender ─▶ mfg/  │
Aug–Oct season,
       Printer test   carte scolaire      (tier counts)    (pre-cleared)   ship/  │
MIP-R1)
       (Phase 0-H)    baseline studies         │                           customs┤
                                               ▼                            (6m)  │
                                          UAT team + ToT ─▶ school training ──────┤
                                          (M8)              + credentialing       │
                                          Print-season contract (QR clauses) ─────┘
```

Calendar-critical items NOT controlled by Opesware: the decree, financing effectiveness, carte scolaire release, the pilot-year print contract, and customs clearance. The plan's discipline: everything Opesware controls finishes early (MIP-R2), every non-controlled item has a named owner, a latest-safe date derived from the SY pin, and — where possible — a fallback (§5 track 4).

8. Gate governance and the failure protocol *(closes gaps #11, #12)*

Operating rhythm: monthly Programme Board (PMU + Opesware + ministry focal points: delivery status vs this plan, risk register deltas, support metrics) · quarterly Steering Committee (gate previews, risk re-scoring per RSK §5, change control) · the **Change Control Board is the Steering Committee in session** — scope additions are only accepted at quarterly reviews and priced against the contingency line (Risk R13).

Gate failure protocol — a gate miss triggers a bounded loop, not a stall:

1. **Diagnose (≤ 4 weeks):** PMU + independent verifier produce a root-cause note per missed indicator (system defect / adoption / data quality / external dependency).
2. **Corrective window (≤ 1 school term):** funded from contingency; corrective-support protocol for adoption misses (MEF §3.3 — support, not sanctions).
3. **Re-test:** the missed indicators only, independently verified.
4. **Second miss → descope decision:** Steering Committee must choose an explicit option — reduce wave size, extend timeline one season (SY-pinned), or activate the module-pause descope plan (Risk R5) — and notify financiers. **Silent target erosion is prohibited; the M&E baseline-freeze rule (MEF §3.2) applies to targets too.**

9. What "done" means (end state, T0+60m)

- Every approved textbook title and new print run nationally identified and passported; ≥90% of circulating stock tracked.
- ≥97% of procured books confirmed received at schools — against a world where 30% of value leaked invisibly.
- 18,500 public schools operating digitally through **two full school years**, including remote schools on rugged tablets syncing by travel.
- Cameroon's first authoritative open school dataset and textbook catalogue, published as national APIs.

- A national team operating the platform — including its helpdesk — on 1.22 bn FCFA/yr, 0.13% of the ministries' budgets.
- The registries ready to carry the next products (equipment passports, inspection, teacher deployment) at marginal cost.

10. v1.0 → v1.1 gap register (traceability)

#	v1.0 gap	Closed by
1	No school-year alignment	§2 MIP-R1 pinning rule; §5 track 4; §6 wave rule
2	Device procurement lead time invisible	§5 track 2 (6-month budgeted lead incl. customs); on the spine §7
3	Real-print-batch dependency unscheduled	§5 track 4 + explicit fallback path
4	Phase 0 duration optimism	§2 MIP-R2 (12–18m financier clock, Opesware finishes in 6)
5	Training cascade untimed	§5 track 3 (ToT M8–M9, waves synced to devices)
6	User provisioning unowned	§5 track 3 (rides data-campaign workshops; ADR-10 creden-tials at training)
7	Publisher/printer onboarding missing	Phase 0-H (label test run) + track 4 contract clauses
8	No support model	§5 track 3: L1 division / L2 PMU helpdesk (within existing bud-get posts) / L3 engineering, live before go-live, with SLAs
9	Security testing unplaced	Pentest 1 gates pilot data (M11); pentest 2 pre-Phase III
10	Ministry UAT capacity absent	§5 track 3: UAT team stood up M8, co-executes from M9
11	No gate-failure protocol	§8: diagnose → corrective window → re-test → explicit de-scope decision
12	No operating rhythm / change control	§8: monthly Board, quarterly Steering = CCB
13	54m vs budget's 60m	Phase IV extended to T0+60m (§3, §6) — matches BUD exactly